

Airline Management System - Sequence Diagrams Reference

This document provides detailed **Sequence Diagrams** (powered by Mermaid) depicting the runtime message flows, synchronous/asynchronous inter-service interactions, authorization handshakes, database transactions, exception paths, and external gateway integrations across the **Airline Management System** microservices platform.

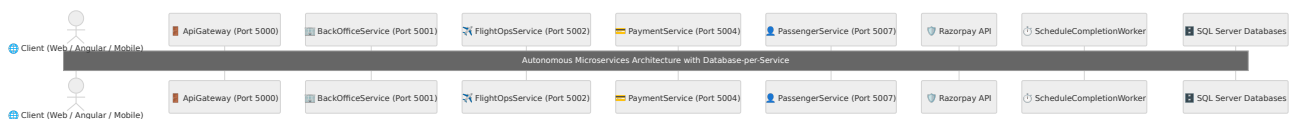
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1. Architecture & Sequence Actors Overview

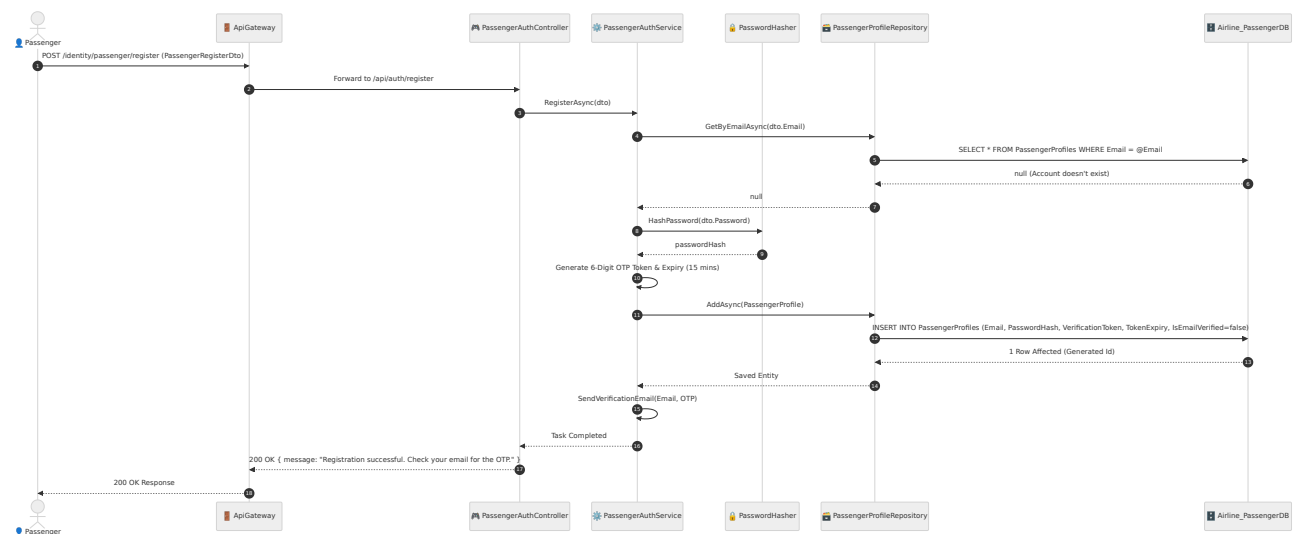
The sequence diagrams model interactions between the following primary participants:



2. Passenger Identity & Access Management

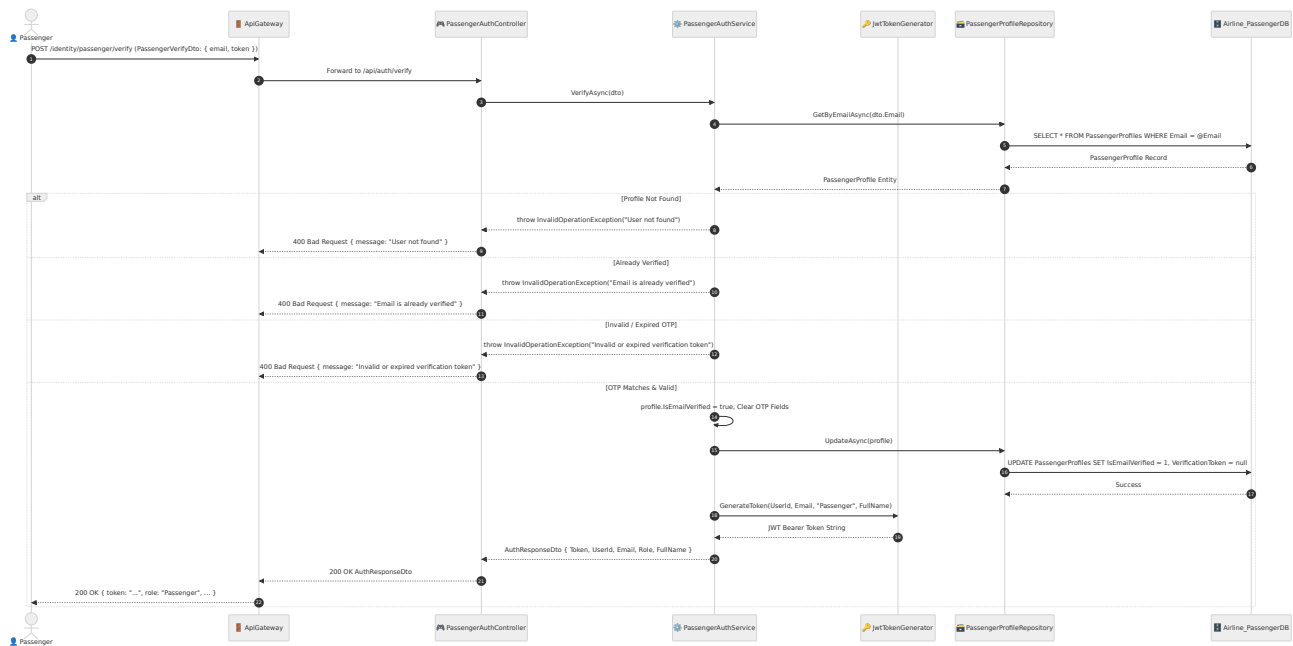
2.1 Passenger Registration & Email Verification OTP Flow

When a new passenger registers on the portal, an account profile is created in an unverified state (`IsEmailVerified = false`), a secure OTP is generated with a 15-minute expiration, and a confirmation email is dispatched.



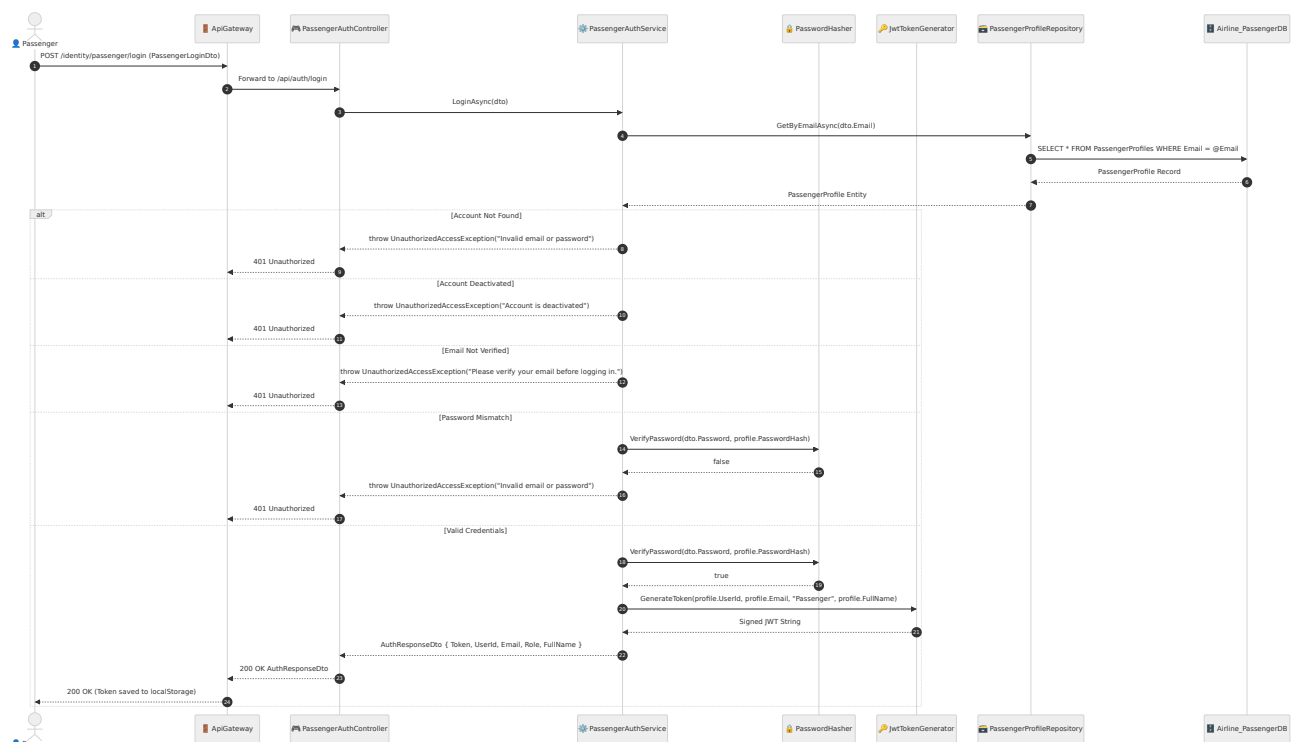
2.2 Passenger Email OTP Verification & JWT Token Issuance

The passenger submits the 6-digit OTP code received in their email to activate the account and receive an initial JWT Bearer authentication token.



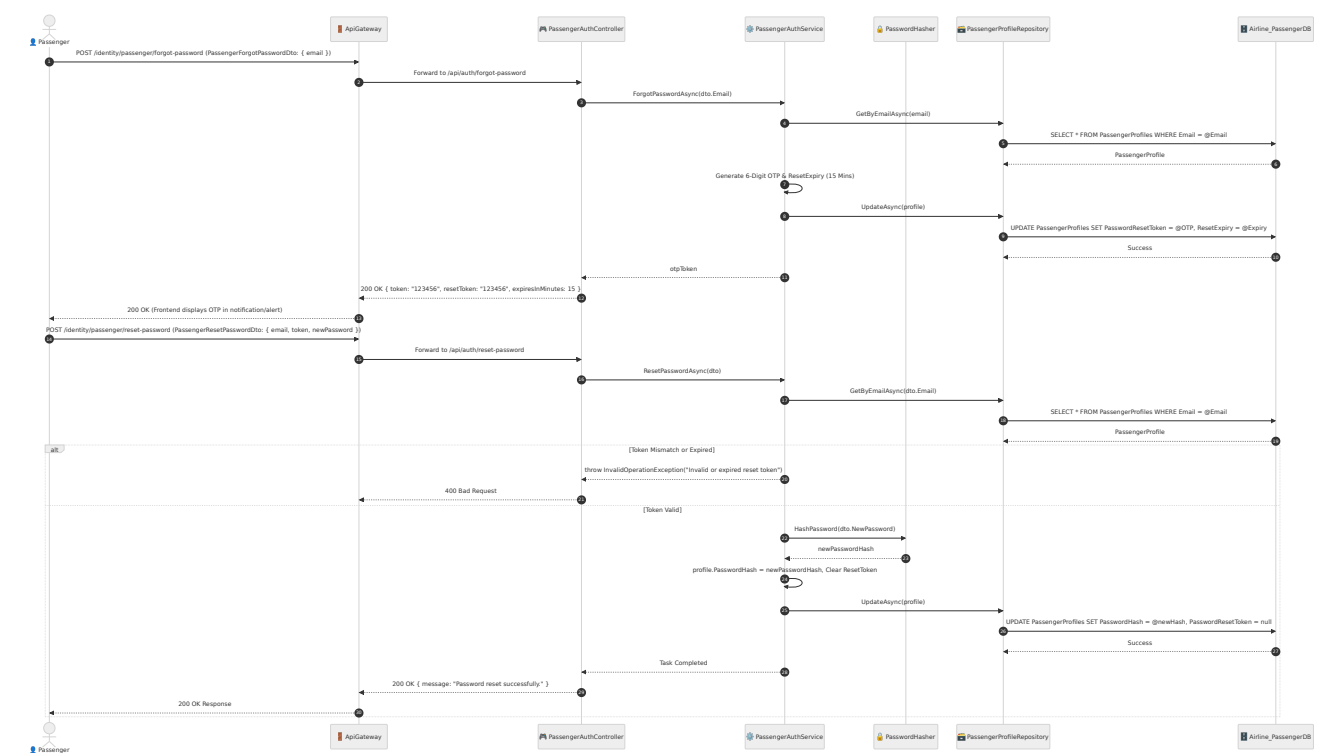
2.3 Passenger Login & Authentication

Existing passengers authenticate with their email and password. The system checks email verification status, verifies the cryptographic password hash, and produces a signed JWT claim token.

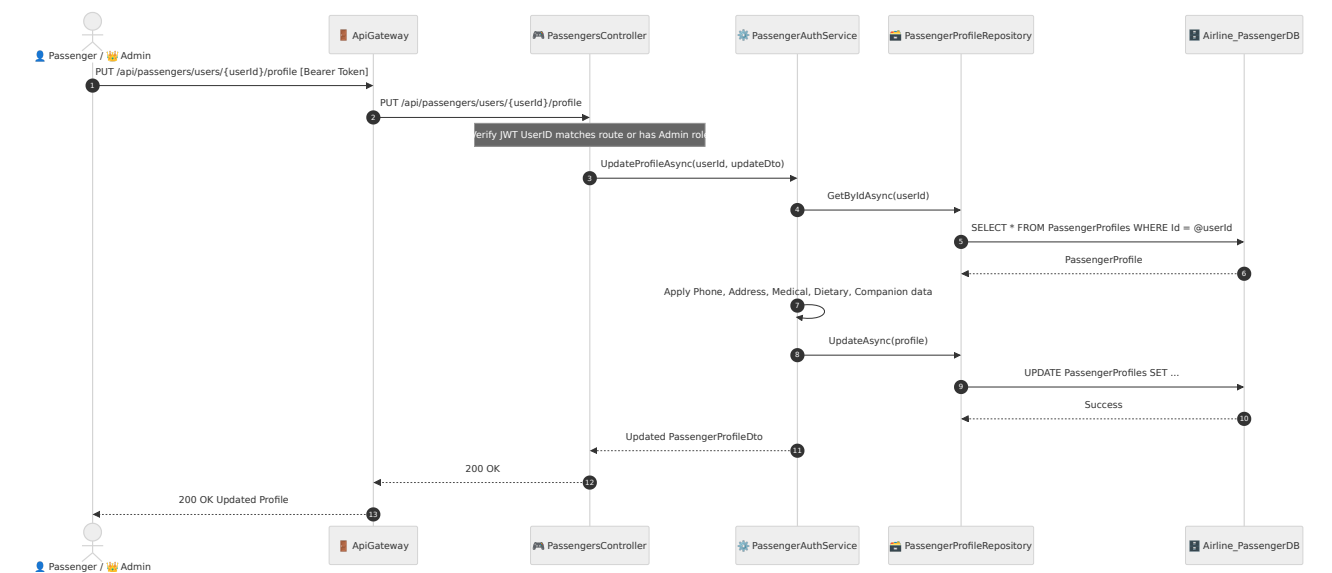


2.4 Passenger Forgot Password & Password Reset Flow

A streamlined self-service password recovery flow that issues a 15-minute reset OTP and verifies the OTP before updating the hashed password.



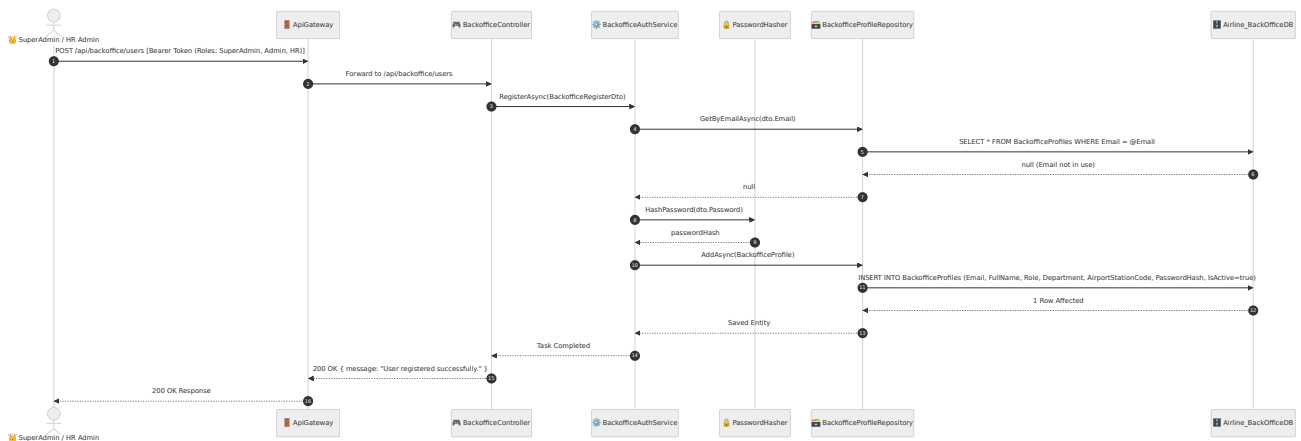
2.5 Passenger Profile Update & Status Management



3. Backoffice Staff Authentication & Administration

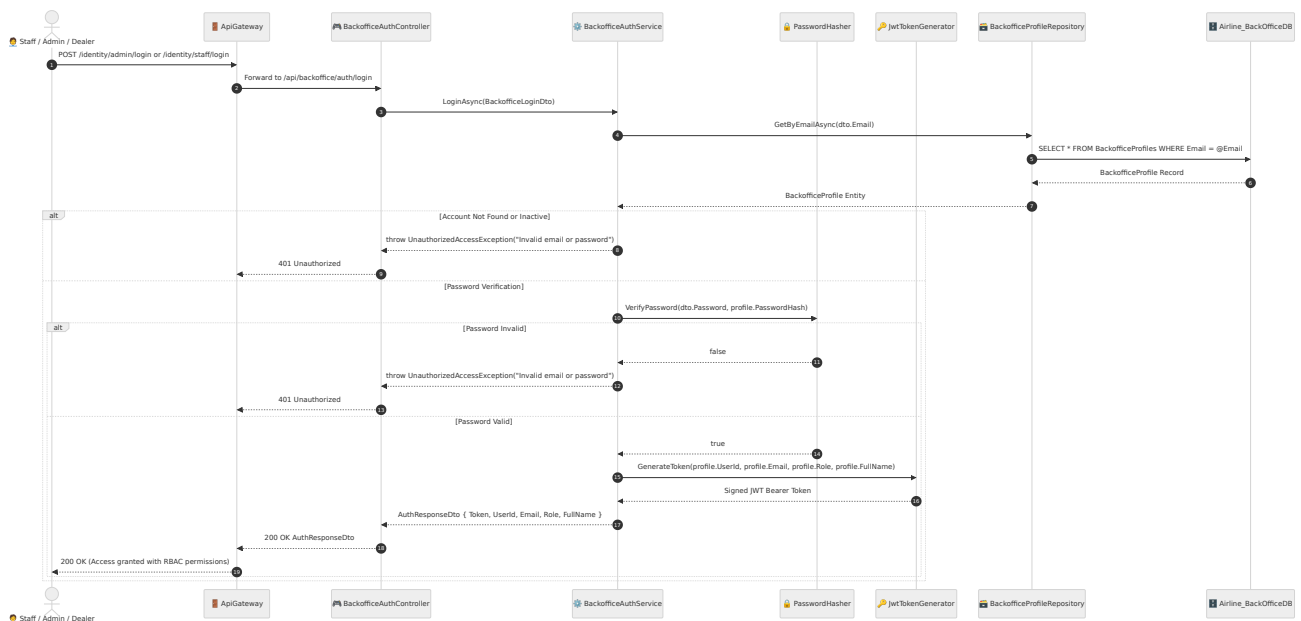
3.1 Backoffice User Provisioning (SuperAdmin / HR / Admin)

SuperAdmins, Admins, and HR managers provision internal staff accounts across all departments (HR, FinancialAdmin, Staff/Crew, GroundStaff, Dealer).

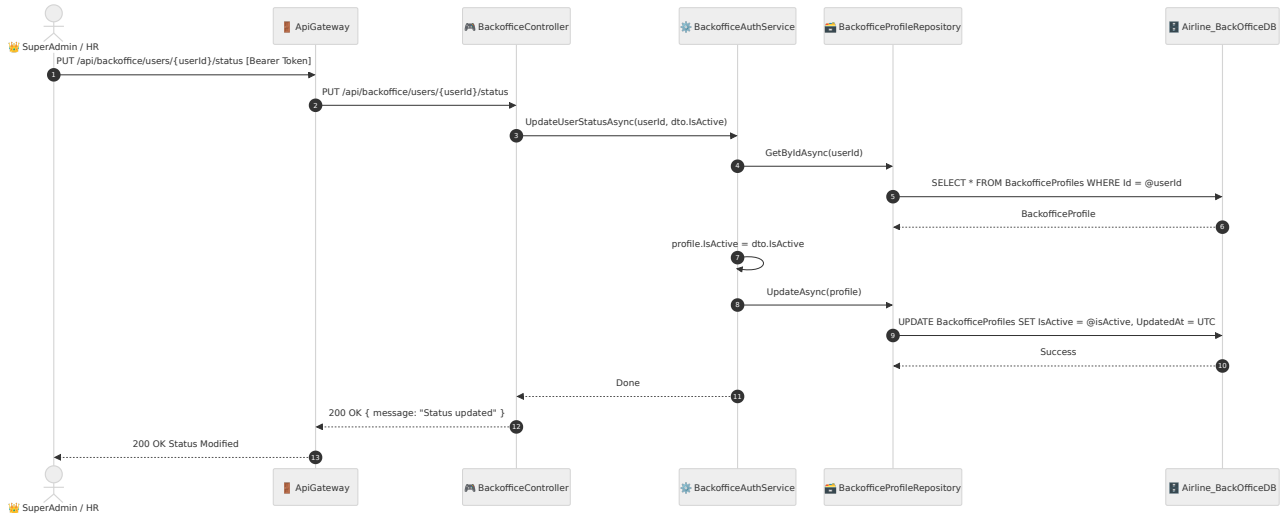


3.2 Backoffice Staff Login & RBAC JWT Issuance

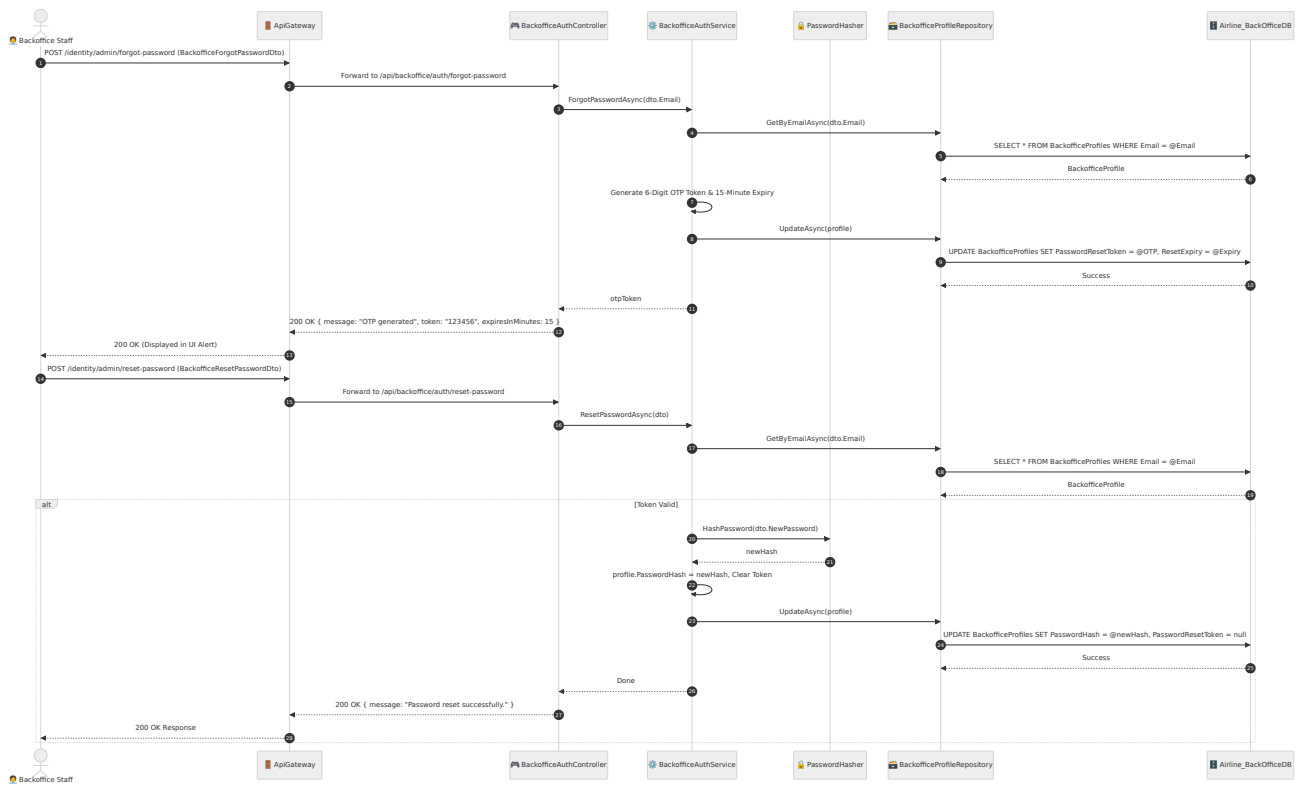
All backoffice personnel log in via a unified portal endpoint (`/api/backoffice/auth/login`), receiving a JWT embedded with their specific administrative role claims.



3.3 Backoffice User Profile & Status Modification



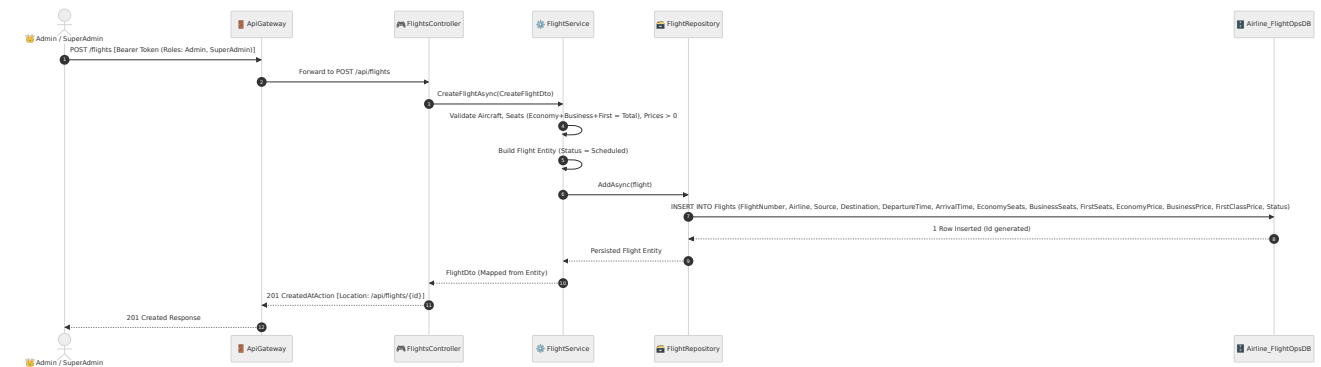
3.4 Backoffice Forgot Password & Password Reset Workflow



4. Flight Catalog & Schedule Operations

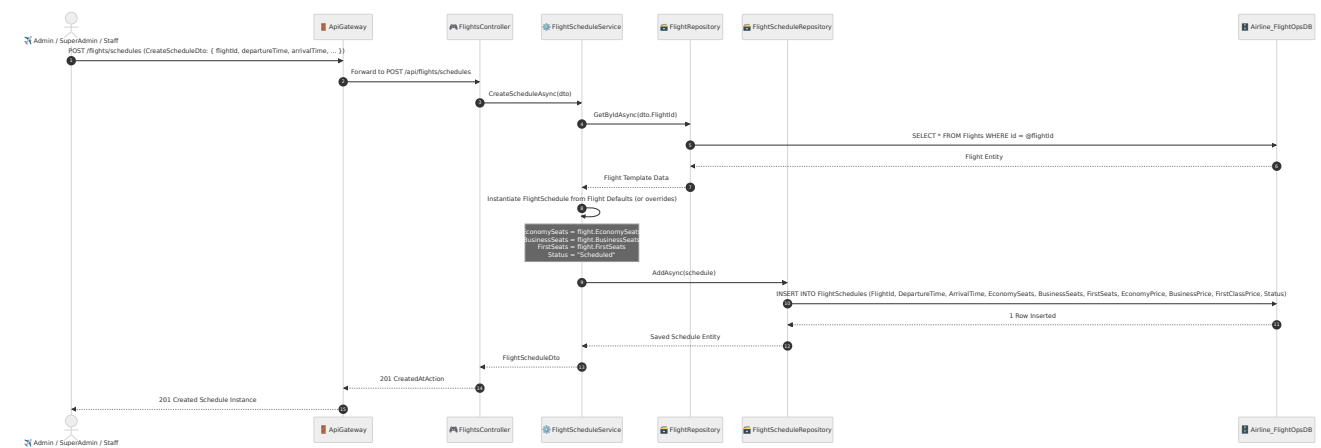
4.1 Flight Template Creation (Admin / SuperAdmin)

Flight templates establish the baseline route, pricing tiers, and cabin class capacities for a given flight number.



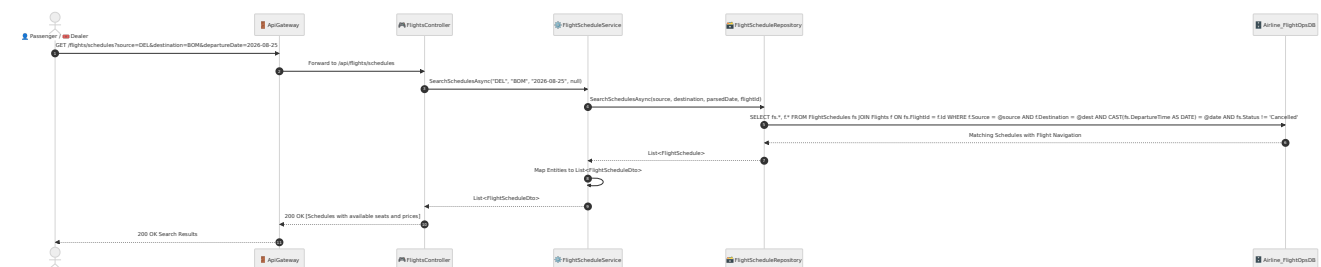
4.2 Flight Schedule Generation & Inventory Setup

Schedules are concrete departure instances derived from flight templates, managing day-to-day seat inventory.

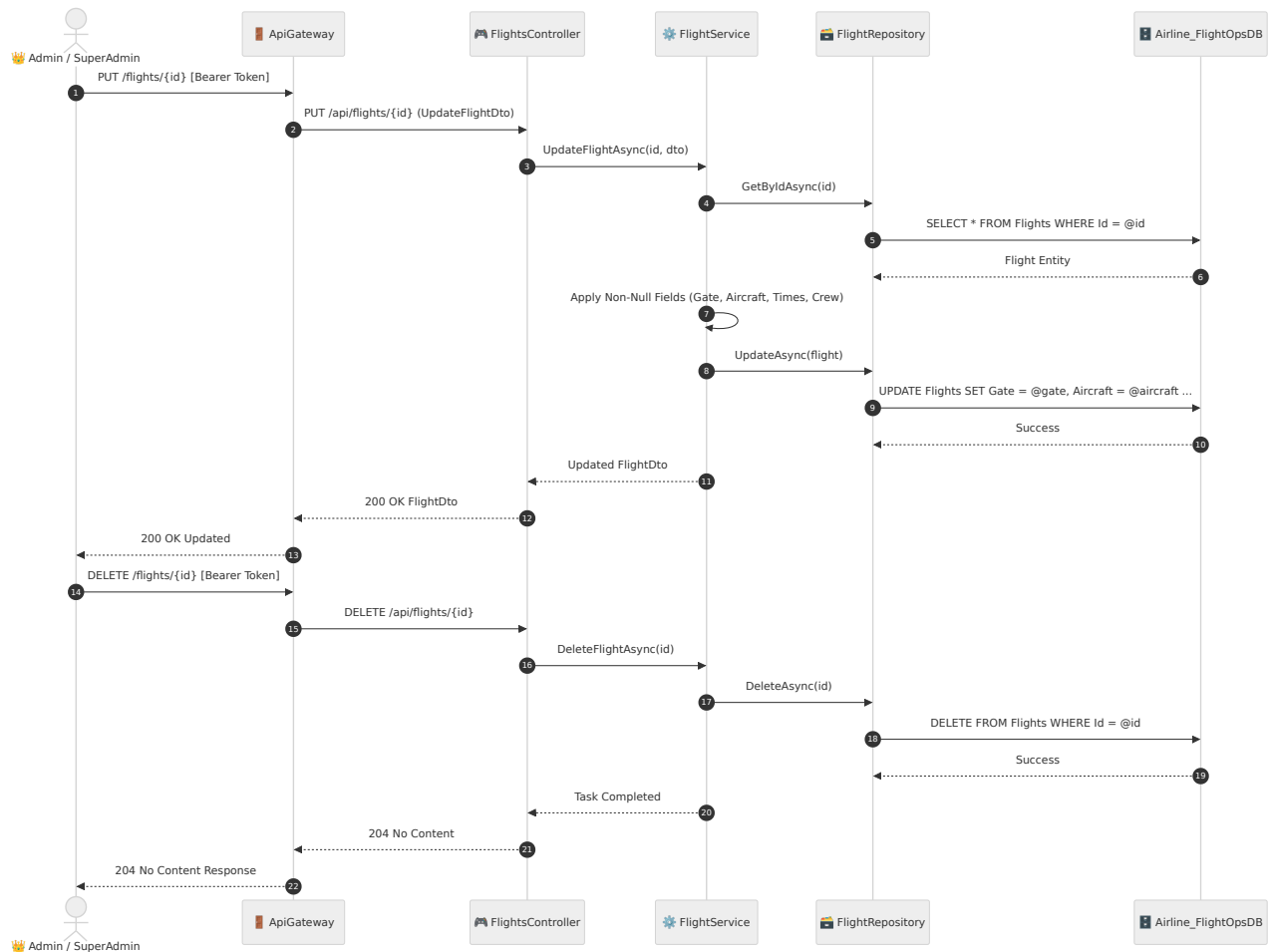


4.3 Public Flight Search & Schedule Query Flow

Passengers and Dealers query flight schedules by origin, destination, and departure date without requiring authentication.



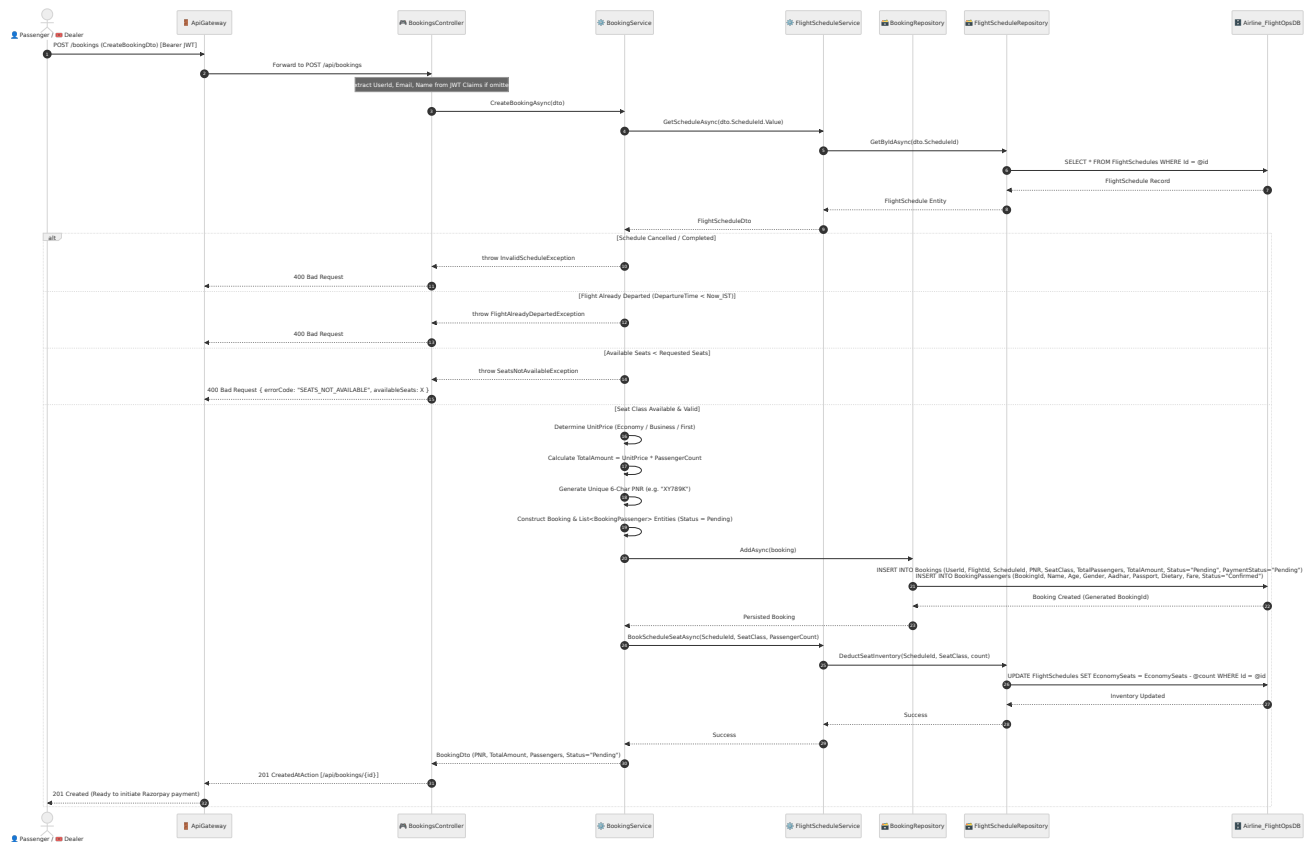
4.4 Flight Template Modification & Removal



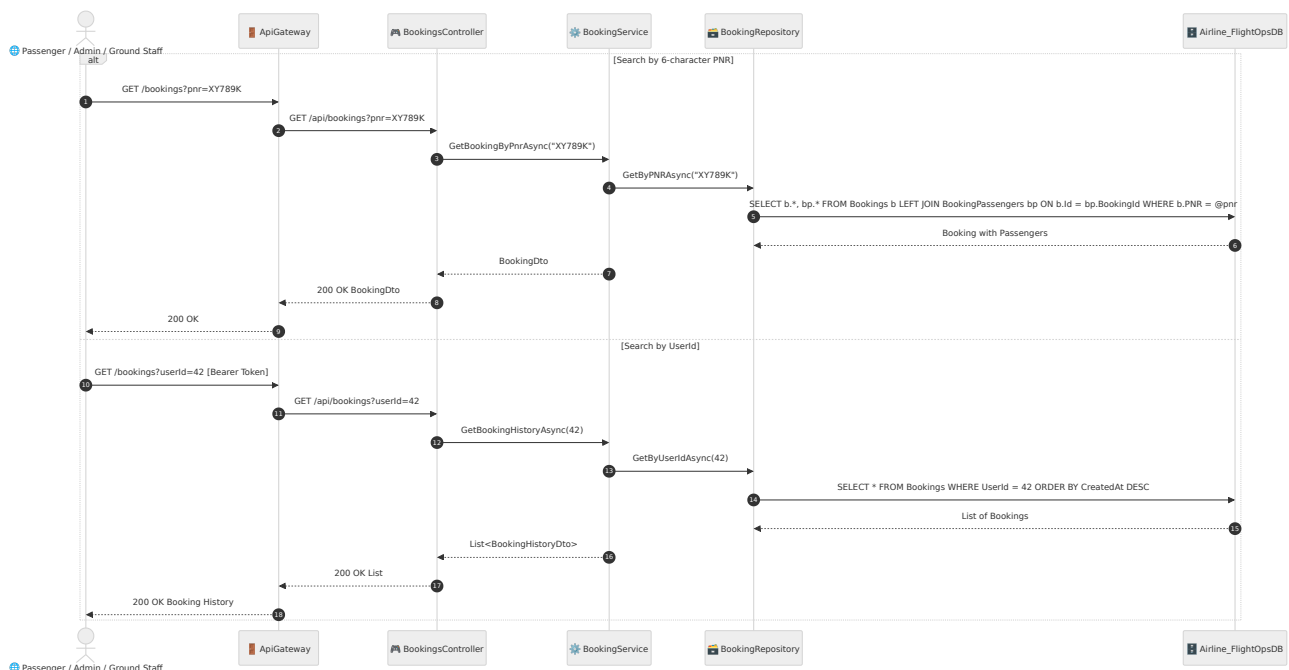
5. End-to-End Flight Booking Lifecycle

5.1 Multi-Passenger Booking Creation, Seat Deduction & PNR Generation

When booking, the system extracts the user identity from JWT claims, verifies seat inventory in-process on the `FlightSchedule` (or `Flight`), deducts seats, generates a unique 6-character PNR, inserts passenger manifest items, and marks the initial state as `Pending` payment.

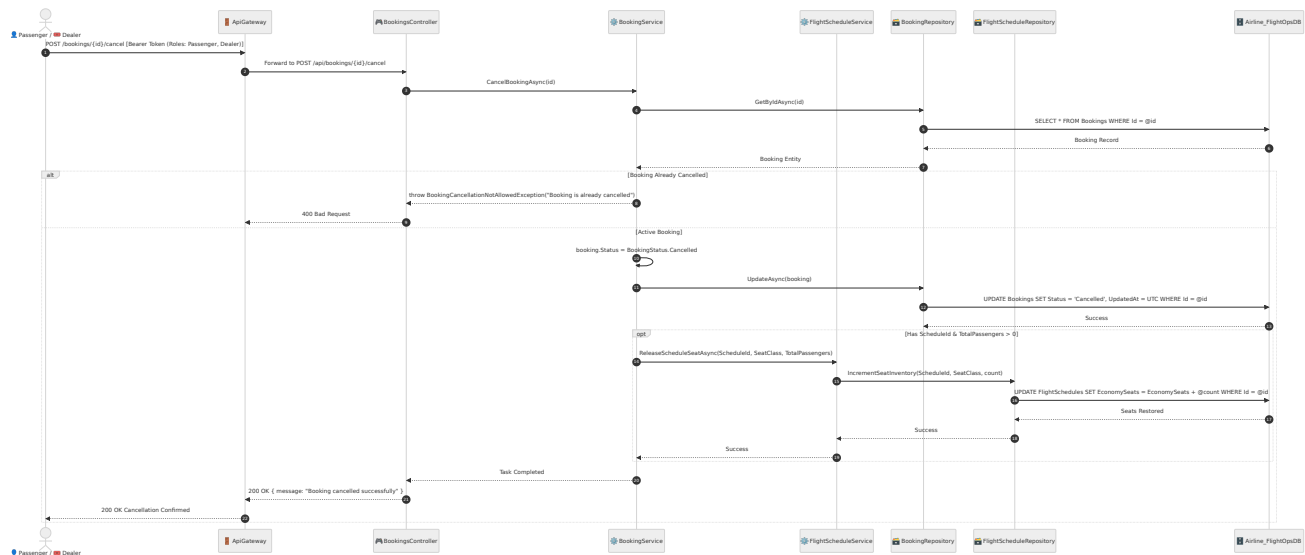


5.2 Booking Inquiries (By PNR, User History, or Schedule)



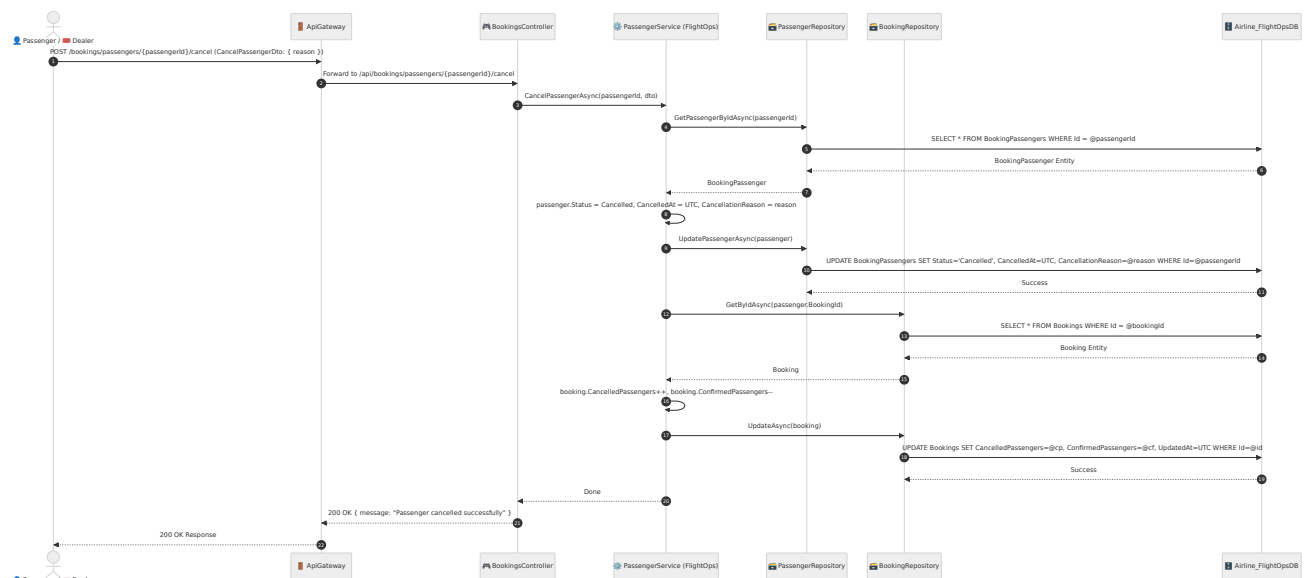
5.3 Full Booking Cancellation & Dynamic Seat Inventory Release

Cancelling a booking updates the booking status to `Cancelled` and automatically restores the reserved seats to the `FlightSchedule` seat inventory.



5.4 Partial Companion Passenger Cancellation

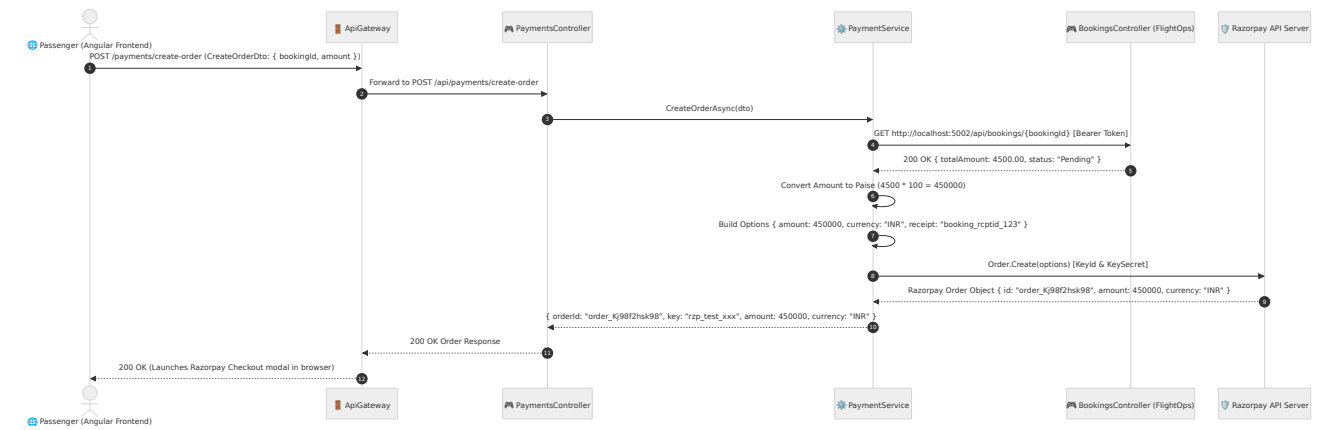
Allows removing an individual companion passenger from a multi-seat booking without cancelling the remaining passengers.



6. Payment Processing & Razorpay Gateway Integration

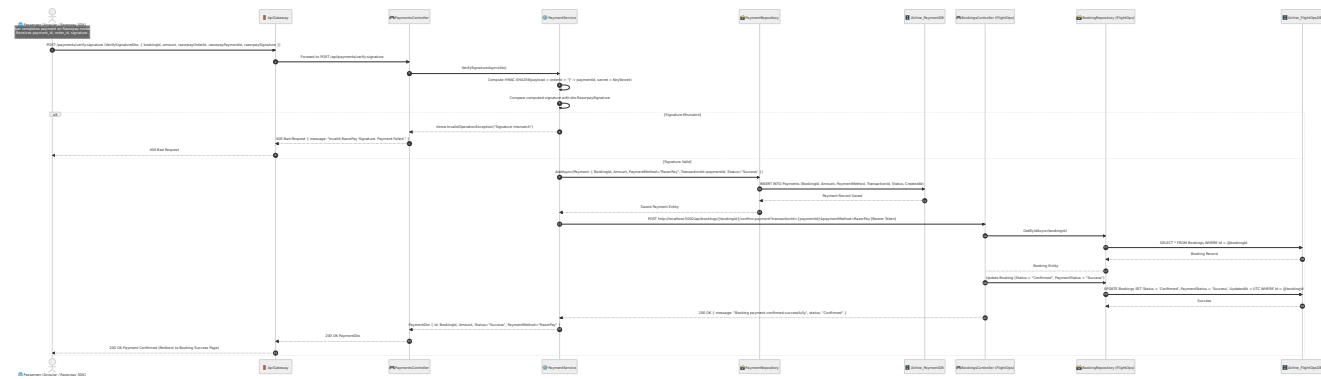
6.1 Razorpay Order Creation Flow

The frontend initiates payment by asking `PaymentService` to create an official Razorpay Order. `PaymentService` calls `FlightOpsService` to validate the booking's authoritative total amount before communicating with Razorpay.



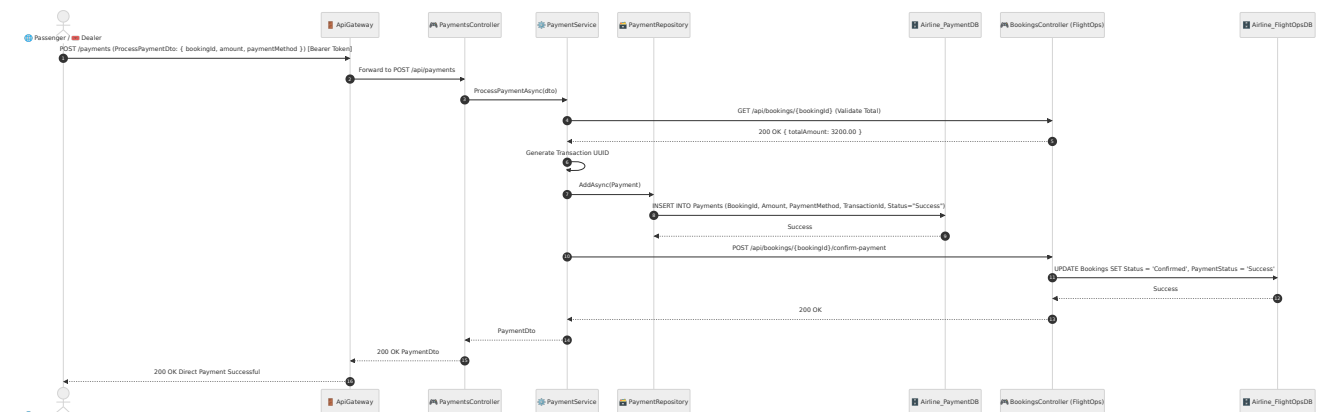
6.2 Razorpay Signature Verification & Cross-Service Booking Confirmation

After the user completes payment on the Razorpay modal, the frontend sends the checkout signatures for cryptographic verification. Once verified, `PaymentService` notifies `FlightOpsService` via REST to confirm the booking.



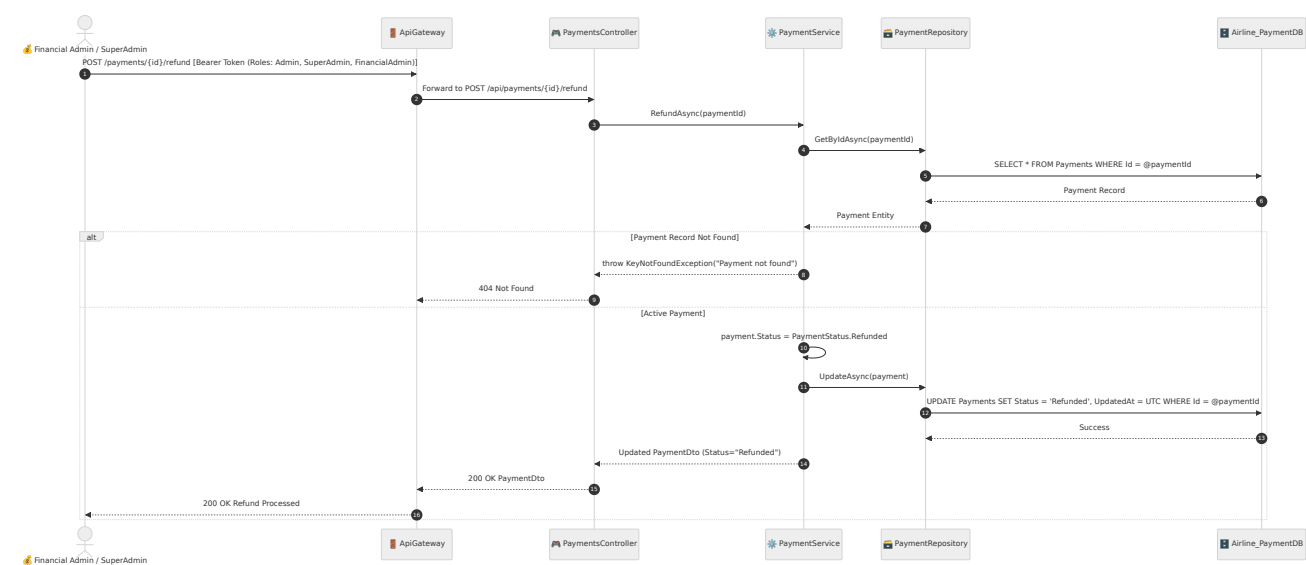
6.3 Direct Payment Processing Workflow

For direct card/UPI simulation endpoints (`POST /api/payments`):



6.4 Payment Refund Processing Flow

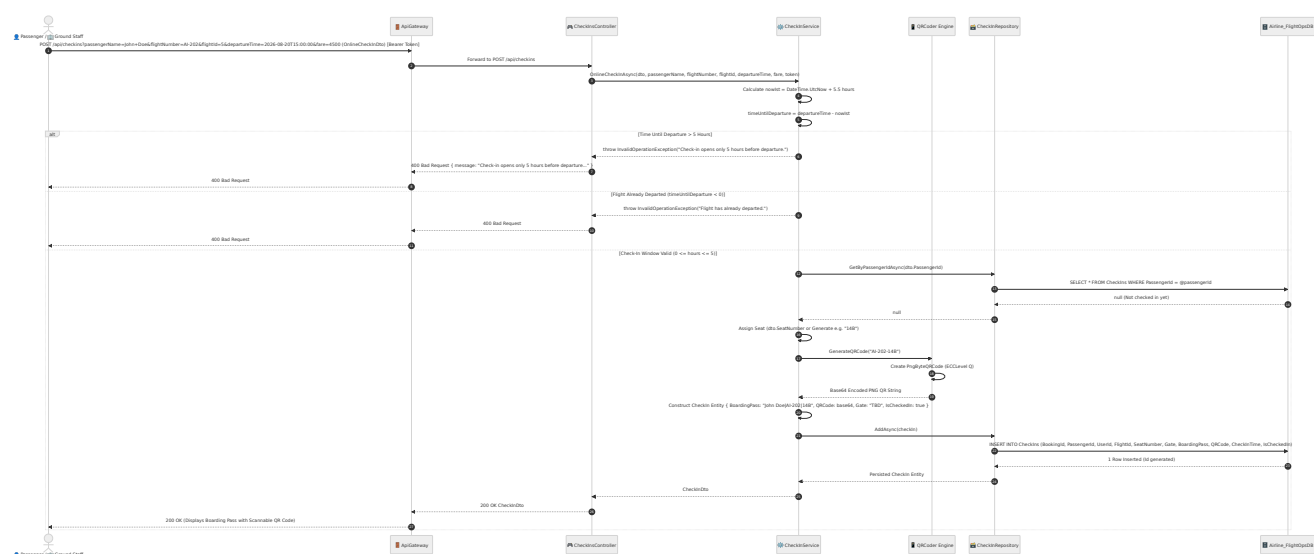
Authorized Financial Admins and SuperAdmins issue refunds for previously successful payments.



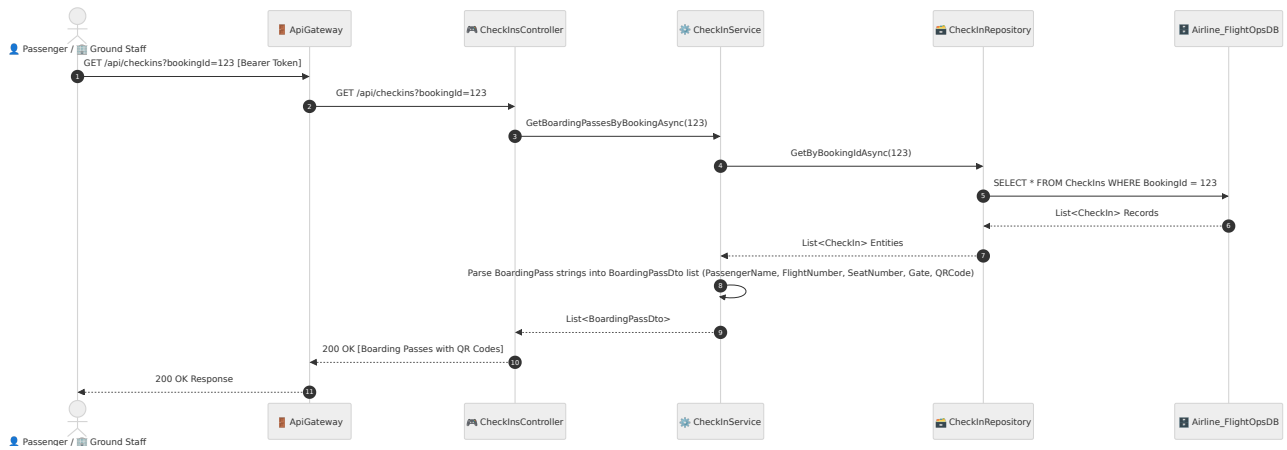
7. Digital Online Check-In & Boarding Pass Issuance

7.1 5-Hour Window Check-In, Seat Assignment & QR Generation

Check-in enforces an IST departure time rule (opens strictly within **5 hours** before departure), generates or assigns the seat number, creates a dynamic Base64 PNG QR code via `QRCode` , and issues the digital boarding pass.



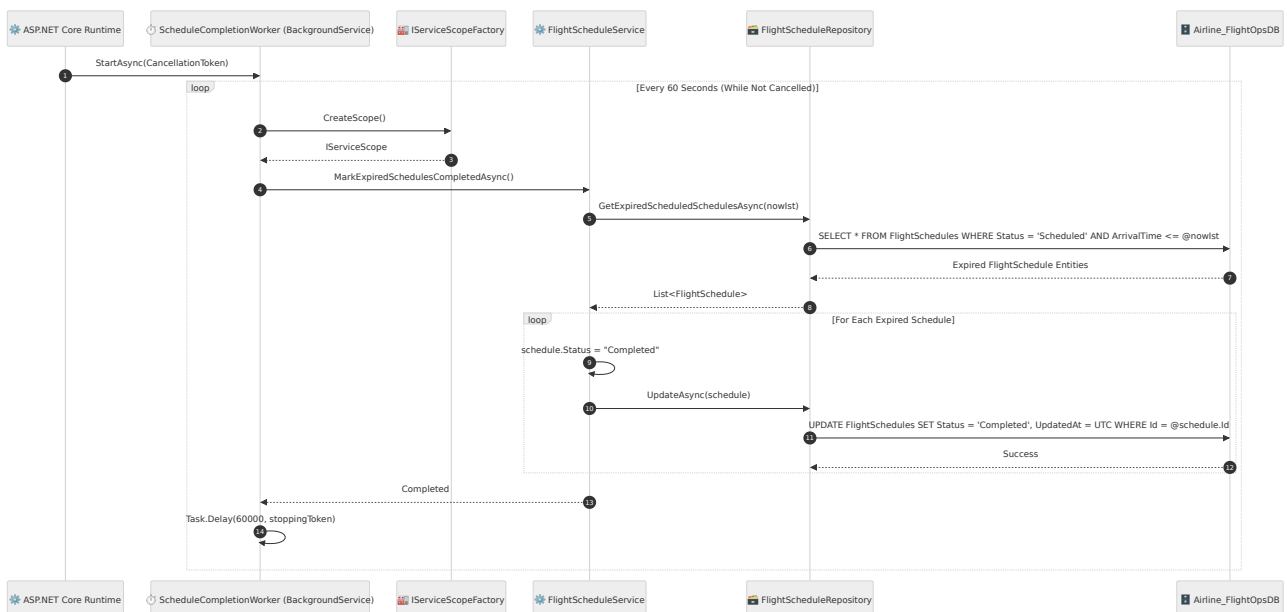
7.2 Fetching Digital Boarding Passes by Booking ID



8. Asynchronous Background Processing

8.1 Autonomous Flight Schedule Auto-Completion Loop

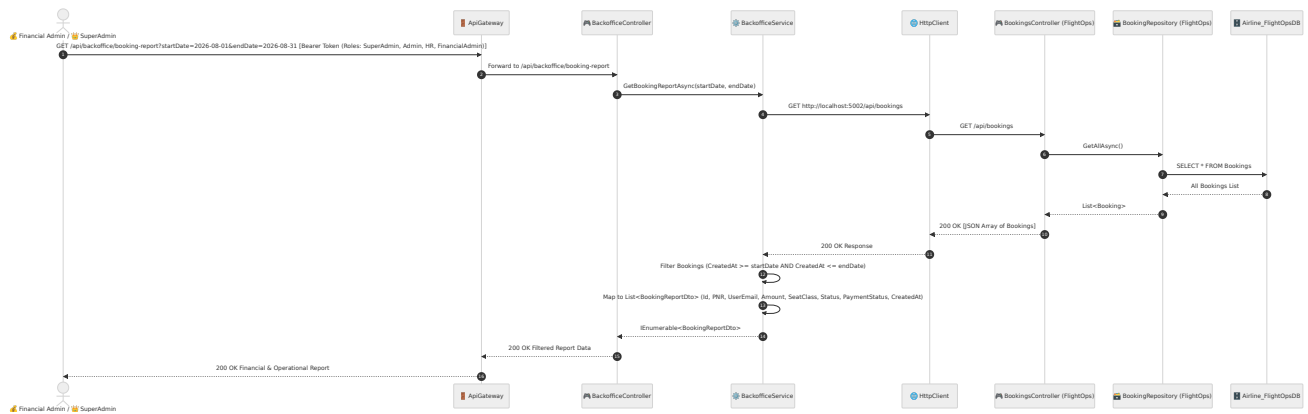
The `ScheduleCompletionWorker` runs as a hosted background service loop every 60 seconds. It identifies flight schedules whose arrival time has elapsed in IST and marks them as `Completed`.



9. Cross-Service Operational & Financial Reporting

9.1 Backoffice Booking & Financial Report Aggregation Flow

The Backoffice Service aggregates financial performance, booking velocity, and occupancy metrics across dates by invoking the `FlightOpsService` API securely over internal HTTP.



Summary of Sequence Interaction Coverage

Feature / Workflow	Primary Initiator	Services & External Components Involved	Outcome / Result
Passenger Registration & OTP	Public Passenger	PassengerService, SQL Server	Profile created (IsEmailVerified=false), 6-digit OTP sent via email.
Email OTP Verification	Passenger	PassengerService, JwtTokenGenerator	Account activated, signed JWT Bearer token issued.
Backoffice Staff Provisioning	SuperAdmin / HR	BackOfficeService, PasswordHasher	Role-based internal staff profile provisioned.
Flight Template & Schedule Creation	Admin / SuperAdmin	FlightOpsService, Airline_FlightOpsDB	Catalog template & schedule instances with inventory generated.
Booking Creation & Seat Deduction	Passenger / Dealer	FlightOpsService (In-Process)	Seats deducted, 6-char PNR created, booking set to Pending.
Razorpay Order Creation	Passenger (Frontend)	PaymentService, FlightOpsService, Razorpay API	Authoritative total validated, order created in paisa.

Feature / Workflow	Primary Initiator	Services & External Components Involved	Outcome / Result
Razorpay Payment Verification	Passenger (Frontend)	PaymentService, FlightOpsService, Razorpay	HMAC-SHA256 signature verified, booking transitioned to Confirmed & Paid.
Digital Online Check-In	Passenger / Ground Staff	FlightOpsService, QRCoder	5-hour window verified, seat assigned, Base64 PNG QR boarding pass issued.
Schedule Auto-Completion Worker	Background Hosted Service	FlightOpsService, SQL Server	Departed flight schedules automatically transitioned to Completed.
Cross-Service Reporting	FinancialAdmin / SuperAdmin	BackOfficeService FlightOpsService	Booking and revenue data aggregated and filtered across date ranges.